Thesis Guideline

The goal of a Master/Bachelor thesis is for the student to perform independent research under the guidance of a suitable supervisor. It is the student's responsibility to proactively adjust their project in response to the challenges inherent in academic research.

1 Thesis Outline

Abstract/Kurzfassung (1 page): If the report is written in English, a German summary must be provided.

1.1 Introduction

Briefly describe your thesis:

- Context: What is our problem?
- Motivation: Why are we interested?
- Objectives: Which aspects do we consider?
- Contribution: How does our work benefit the scientific community?
- Outline: Where in the report will we describe what?

Give a detailed summary of the existing literature; describe which issues have been addressed (and how) and where existing solutions have been limited. Clearly describe the "gap" in the literature that motivates your work.

1.2 Theoretical background

Formulate the problem mathematically. What do we already know about it? Give and explain applicable theoretical results; be careful when citing theorems and/or proofs from the literature. Be sure to include sufficient mathematical background for the ideas you build on, e.g., model-predictive control, reference governors, invariant sets, etc.

1.3 Main results

Document what you have done:

- What are the assumptions (e.g., dynamic model)?
- If you did experiments or simulations, what is the set-up?
- What are the advantages of your approach, what are the limitations?
- What results did you obtain?

Be sure to analyze your results and put it into context of previous results! If you plan on presenting the results in a scientific conference or journal, include your paper here. Regardless of submission requirements, please choose a single-column layout and continue the pagination from the previous sections (or remove page numbers altogether).

Otherwise, a suitable outline for the report could be (at section or subsection level):

- Experimental set-up (if applicable)
- Analysis/Results
- Discussion

1.4 Conclusions & Future work

Briefly summarize the preliminaries, your assumptions, the work performed, and the results obtained. How can the results be interpreted and what are further insights? Describe how the limitations of this work could be resoved and/or identify possible research directions building up on your results.

1.5 Bibliography

List all references that you have made into existing literature both in the report and in your paper (if applicable)! Appropriate citation styles are: numerical, alpha-numeric, or author- year.

1.6 Appendix

Provide further details on your set-up, results, and/or the experiments you have performed. If you presented your main results as a paper, this is mandatory!

2 Tools

1. The standard tool for writing scientific documents is `LaTex`. It is a typesetting system that is widely used in academia for its flexibility and professional appearance. You can find a comprehensive guide to LaTex at `https://www.latex-project.org/`. In addition, the thesis template provided is only in LaTex format.
2. To write your LaTex document, you can use an editor like `Texmaker`, `Overleaf`, or `TeXShop`. These editors provide a user-friendly interface for writing LaTex code and compiling it into a PDF document. The software can be found at `https://www.xm1math.net/texmaker/`, `https://www.overleaf.com/`, and `http://pages.uoregon.edu/koch/texshop/`. From a personal preference, we recomment using `VSCode` with the `LaTex Workshop` extension.
3. To keep track of your changes, you can use a version control system like `Git` and a hosting service like `GitHub`. These tools allow you to track changes to your documents, revert to previous versions, and collaborate with others on the same project. The software can be found at `https://git-scm.com/` and `https://github.com/`.
4. To create figures in Matlab and easily include them in your LaTex document, you can use the `matlab2tikz` package. This package converts your Matlab figures to TikZ code, which can be included in your LaTex document. The package can be found at `https://github.com/matlab2tikz/matlab2tikz`.
5. To keep track of your references, you can use a reference manager like `Zotero`, `Mendeley`, or `Endnote`. These tools allow you to easily import references from online databases, organize them in folders, and automatically generate a bibliography in your LaTex document. From a personal preference, we recommend using `Zotero` as it is free and open-source. The software can be found at `https://www.zotero.org/`.

3 Notation and Style

Due to the lack of experience from the majority of students in writing scientific documents, we provide a list of common mistakes and suggestions to improve the quality of your thesis, focusing more on the mathematical notation and LaTex writing.

a. **Common Abbreviations**

Make sure to use the correct abbreviations and symbols in your document. Some common abbreviations and their meanings are:

- **e.g.** (exempli gratia) is used to introduce an example.
- **i.e.** (id est) is used to clarify or explain a statement.
- **w.r.t.** (with respect to) is used to indicate the subject of a comparison.
- **et al.** (et alii) is used to indicate multiple authors.
- **Fig.** is used to refer to a figure.
- **Eq.** is used to refer to an equation.
- **Sec.** is used to refer to a section.
- **Tab.** is used to refer to a table.

See, for example, the IEEE Editorial Style Manual for more abbreviations.

b. **In-Line expressions**:

- Fractions must not appear stacked in in-line expressions. Use a slash instead.
  For example, write a/b instead of $\frac{a}{b}$.
- Use parentheses to clarify the order of operations.
  For example, write $a/(b+c)$ instead of $a/b+c$.
- Collective signs such as $\sum$, $\prod$, $\int$, etc., should not have the limits at the bottom and top, but instead to the right.
  For example, you should write $\sum_{i=1}^{n} a_i$.

c. **Theorems, Proofs and Definitions**:

To write theorems, proofs, and definitions, you can use the amsthm package in LaTeX. This package provides a set of environments for writing mathematical content. For example, you can write a theorem as follows:

```
\newtheorem{theorem}{Theorem}
\begin{theorem}
  Let $a$ and $b$ be real numbers. Then $a + b = b + a$.
\end{theorem}
```

This will produce the following output:

Theorem 1 *Let a and b be real numbers. Then $a + b = b + a$.*

To create your own theorem environment (e.g. Remarks), you can use the following command:

```
\newtheorem{remark}{Remark}
\begin{remark}
  The sum of two real numbers is commutative.
\end{remark}
```

This will produce the following output:

Remark 1 *The sum of two real numbers is commutative.*

d. **Equations**:

When referring to equations, use the command eqref instead of ref. This will automatically add parentheses around the equation number. For example, you can refer to an equation as follows:

```
Equation \eqref{eq:example} shows an example of an equation.
\begin{equation}
  \label{eq:example}
  y = mx + b
\end{equation}
```

This will produce the following output:

Equation (1) shows an example of an equation.

$$y = mx + b \tag{1}$$

IMPORTANT: Number only the important equations that are referenced later in the text. Too many numbers can clutter the page. You do not need to number ALL equations!

If you have multi-line equations, align them properly to enhance readability. Use align or split environments (in LaTeX) to organize them.

```
\begin{align}
  y &= mx + b \\
  &= \frac{a}{b} + c
\end{align}

\begin{equation}
  \begin{split}
    y &= mx + b \\
    &= \frac{a}{b} + c
  \end{split}
\end{equation}
```

This will produce the following output:

$$
\begin{aligned}
y &= mx + b \qquad (2)\\
&= \frac{a}{b} + c \qquad (3)
\end{aligned}
$$

$$
\begin{split}
y &= mx + b \\
&= \frac{a}{b} + c
\end{split} \qquad (4)
$$

e. **Reminders**:

Functions and operators must be set in Roman font! For example, use $\sin(x)$ instead of $sin(x)$.

If you have an operator that is not predefined in LaTex, you can define it yourself. For example, to define the operator $\arg\min$, you can use the following command that must be included in the preamble:

```
\DeclareMathOperator*{\argmin}{arg\,min}
```

Note that the package `amsmath` must be included in the preamble. Alternatively, you can use

```
\operatorname{argmin}
```

to write the operator in the same way. (this is useful if you do not want the define in the preamble the operator)

When dealing with subscripts, we need to be careful about what is an index and what is a name of a variable. For example, if we have a variable x and we want to refer to its i-th element, we should write x_i instead of x_i. But if we have the variable x_{real} and we want to refer to its i-th element, we should write $x_{\mathrm{real},i}$. Note that the name is in Roman font and the index is in italic font!

f. **Figures and Tables**:

Figures and tables should be referenced in the text and have a caption that describes the content. For example, you can refer to a figure as follows:

```
Figure \ref{fig:example} shows an example of a figure.
```

IMPORTANT: the caption of the figure should be placed below the figure, while the caption of the table should be placed above the table!

Creating appealing tables can be challenging, we recomment check the guide from Markus Püschel at `https://people.inf.ethz.ch/markusp/teaching/guides/guide-tables.pdf`.

But for a simple example of what a bad and a good table looks like, see below:

Table 1: Not a good example of a table.

Column 1	**Column 2**	**Column 3**
Row 1	1	2
Row 2	3	4
Row 3	5	6

Table 2: A better example of a table.

Column 1	**Column 2**	**Column 3**
Row 1	1	2
Row 2	3	4
Row 3	5	6

g. **Mathematical notation**:

Here are some tips for writing mathematical notation:

- General sets: $\mathbb{R}$ for the real numbers, $\mathbb{N}$ for the natural numbers, $\mathbb{Z}$ for the integers, $\mathbb{Q}$ for the rational numbers, $\mathbb{C}$ for the complex numbers. (use `mathbb` for general sets)
- User-defined sets: $\mathcal{A}$, $\mathcal{B}$, $\mathcal{C}$, $\mathcal{X}$, $\mathcal{Y}$, $\mathcal{Z}$. (use `mathcal` for user-defined sets)
- New notation: if you introduce a new notation, make sure to define it in the text or in the notation section! For example, if you define a new set $\mathcal{S}$, you should write: "Let $\mathcal{S}$ be a set of elements that satisfy the following properties: ..."
- Quantifiers: Use $\forall$ and $\exists$ for "for all" and "there exists", respectively. Even if you use the English words, these symbols have a correct way of being used, namely they should appear before the statement. For example, write $\forall x \in \mathbb{R}\ P(x) \Rightarrow Q(x)$ instead of $P(x) \Rightarrow Q(x)\ \forall x \in \mathbb{R}$. Thus, be careful with the order of the quantifiers and do not use them as a more compact way of writing the statement.

The most important thing is to be consistent in the notation you use throughout your document. Try not to mix different notations for the same concept.